

Experiment 2 – Analog-to-Digital Converter (Flash-and-Half ADC)

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Abstract—This report describes the design and implementation of a Flash-and-Half Analog-to-Digital Converter (ADC) using discrete components on a breadboard. The converter is based on a resistive voltage divider, four comparators and a 4-to-2 priority encoder, providing a 2-bit digital representation of an analog input generated by a potentiometer. The static behavior of the ADC is observed using LEDs, while the dynamic response is evaluated using an oscilloscope for several input frequencies. The experimental waveforms confirm correct digital conversion and demonstrate the effect of input frequency on the quality of the output signal.

Index Terms—Analog-to-digital converter, flash ADC, comparators, priority encoder, breadboard implementation.

I. INTRODUCTION

Many real-world signals such as sound, temperature and sensor outputs are inherently analog, whereas modern digital systems operate on binary data. An Analog-to-Digital Converter (ADC) bridges this gap by mapping a continuous input voltage into a discrete digital code.

There exist several ADC architectures including Flash, Successive Approximation Register (SAR), Sigma-Delta and Pipeline converters. In this experiment we implement a small Flash-and-Half ADC, a hybrid architecture that combines the high speed of a Flash ADC with the hardware efficiency of SAR-like approaches. The implemented circuit converts the voltage of a 10 k Ω potentiometer into a 2-bit digital output that can be used in later experiments.

II. THEORY OF OPERATION

The core of the implemented ADC is shown conceptually in the laboratory manual [1]. A set of five equal resistors forms a voltage divider between the supply rails, generating four reference threshold voltages. Four comparators compare the analog input voltage V_{in} with these thresholds.

For each comparator i , the output is ideally given by

$$V_{o,i} = \begin{cases} \text{HIGH}, & V_{in} > V_{ref,i}, \\ \text{LOW}, & V_{in} \leq V_{ref,i}. \end{cases} \quad (1)$$

The four comparator outputs form a thermometer-like code: as the input voltage increases, a larger number of comparators switch to HIGH.

Because the priority encoder IC 74HC148 is active low, a 7404 inverter is used so that the encoder receives active-low inputs as required. The 74HC148 then converts the 4-line input into a 2-bit binary output. The resulting digital code changes whenever the analog input crosses one of the reference thresholds. Two LEDs connected to the output bits provide an immediate visual indication of the code.

Assuming an input range from 0 to V_{CC} and $N = 2$ output bits, the ADC has $2^N = 4$ quantization levels. The ideal voltage step, or least significant bit (LSB), is therefore

$$\Delta V = \frac{V_{CC}}{4}. \quad (2)$$

III. EXPERIMENTAL SETUP

The circuit was assembled on a solderless breadboard using discrete components, as required by the lab instruction. Figures 1, 2, 3 and 4 show different views of the final implementation, including the comparators, the 74HC148 priority encoder, the 7404 inverter and the 10 k Ω potentiometer that generates the input voltage.

A DC supply provided V_{CC} , and the input voltage was varied by rotating the potentiometer. The digital outputs of the encoder were observed both with LEDs and on an oscilloscope. To study the dynamic response, the input node (or one internal node of the ADC) was also driven by a periodic signal and its corresponding digital output waveform was captured at several frequencies.

IV. RESULTS AND DISCUSSION

A. Static Behavior

With the input connected only to the potentiometer, four distinct regions of operation were observed as the potentiometer wiper was rotated from the minimum to the maximum voltage:

- For the lowest range of V_{in} , both LEDs were OFF, indicating the binary code “00”.
- As V_{in} crossed the first threshold of the resistor ladder, the least significant bit (LSB) LED turned ON while the most significant bit (MSB) remained OFF, corresponding to code “01”.
- Increasing V_{in} further produced the code “10”, where only the MSB LED was ON.

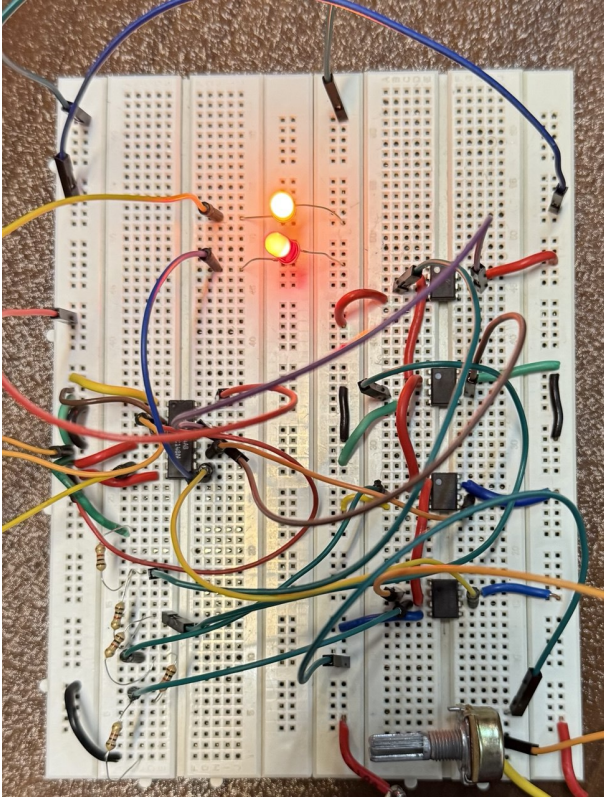


Fig. 1. Overall view of the Flash-and-Half ADC implemented on a breadboard. Two LEDs display the 2-bit digital output.

- For the highest range of V_{in} , both LEDs were ON, corresponding to code “11”.

These observations confirm the correct qualitative behavior of a 2-bit ADC with four quantization regions.

B. Dynamic Waveforms

To investigate the timing behavior and signal integrity of the ADC, several oscilloscope screenshots were recorded. Each screenshot shows a digital waveform at the output of the circuit together with the measured fundamental frequency.

Figure 5 shows a nearly symmetric square wave with a measured frequency of approximately 175.6 kHz. The duty cycle is close to 50 %, indicating that the digital output is correctly following a periodic excitation at this rate.

In Fig. 6, the frequency is reduced to approximately 64.9 kHz. The transitions are slower relative to the period, and the waveform shows less high-frequency ringing. This confirms that the comparator and encoder can comfortably operate in this frequency range.

When the excitation frequency is increased to the megahertz range, as in Fig. 7, the oscilloscope reports a fundamental frequency around 1.35 MHz. At this rate, noticeable ringing and overshoot appear after each transition. These effects are mainly due to the limited slew rate and bandwidth of the op-amp based comparators, as well as parasitic capacitances and wiring inductances on the breadboard.

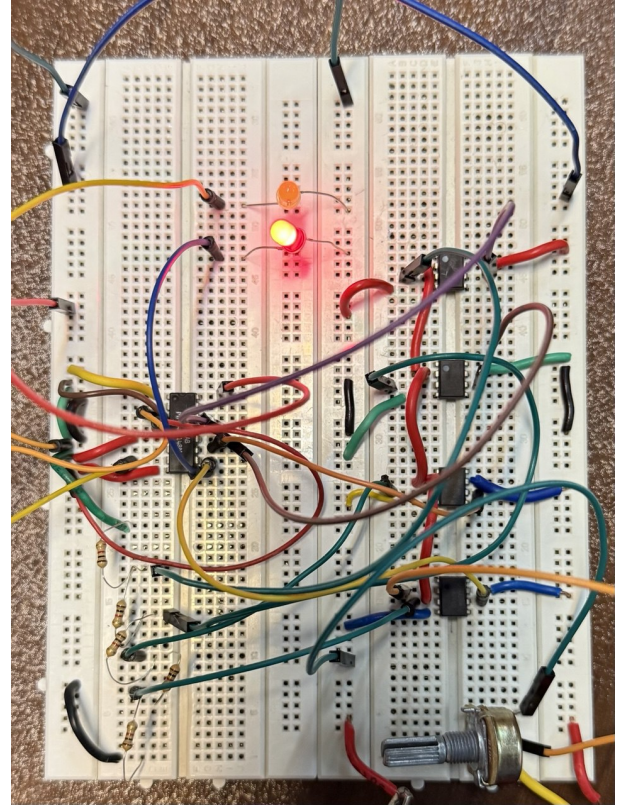


Fig. 2. Breadboard implementation when the analog input is adjusted to a different voltage level, producing another output code on the LEDs.

Finally, Fig. 8 presents another measurement in the hundreds of kilohertz range, with a reported frequency close to 173.2 kHz. The waveform is again a clean square wave, confirming that this range is within the comfortable operating region of the implemented Flash-and-Half ADC.

Overall, the experimental results show that the circuit behaves ideally at tens to hundreds of kilohertz, while at frequencies above 1 MHz non-ideal analog effects become significant. This observation is consistent with the expected limitations of the LM393/UA741 comparator devices and the breadboard wiring.

V. CONCLUSION

In this experiment, a simple Flash-and-Half ADC was designed and implemented using discrete comparators, a resistor ladder and a priority encoder. The converter successfully mapped the analog voltage of a potentiometer into a 2-bit digital code, which was visualized by LEDs and verified with an oscilloscope.

The static measurements confirmed the correct ordering of quantization regions, and the dynamic tests demonstrated reliable operation up to the hundreds of kilohertz range. At higher frequencies around 1.35 MHz, ringing and waveform distortion were observed, highlighting the practical bandwidth limitations of real analog components and of breadboard implementations.

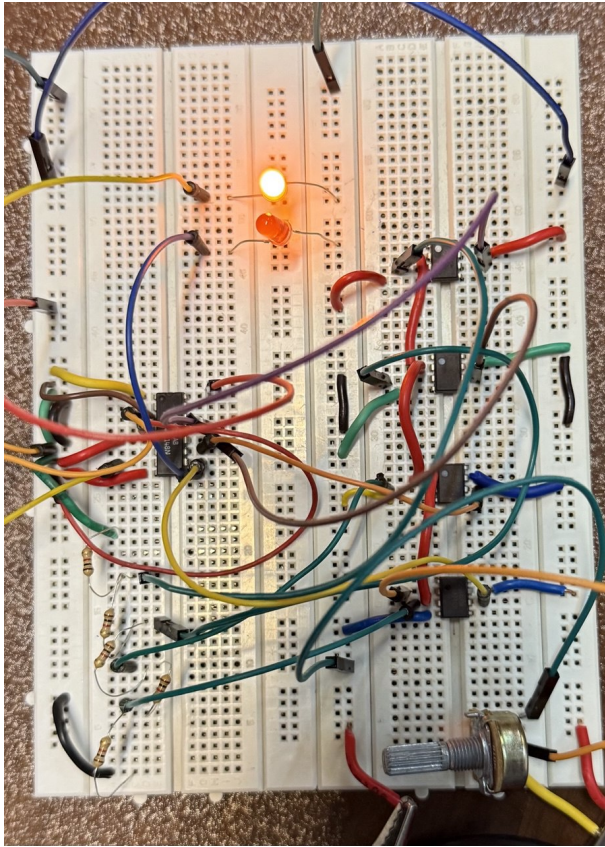


Fig. 3. Close-up of the digital part of the circuit, including the 74HC148 priority encoder and the 7404 inverters.

ACKNOWLEDGMENT

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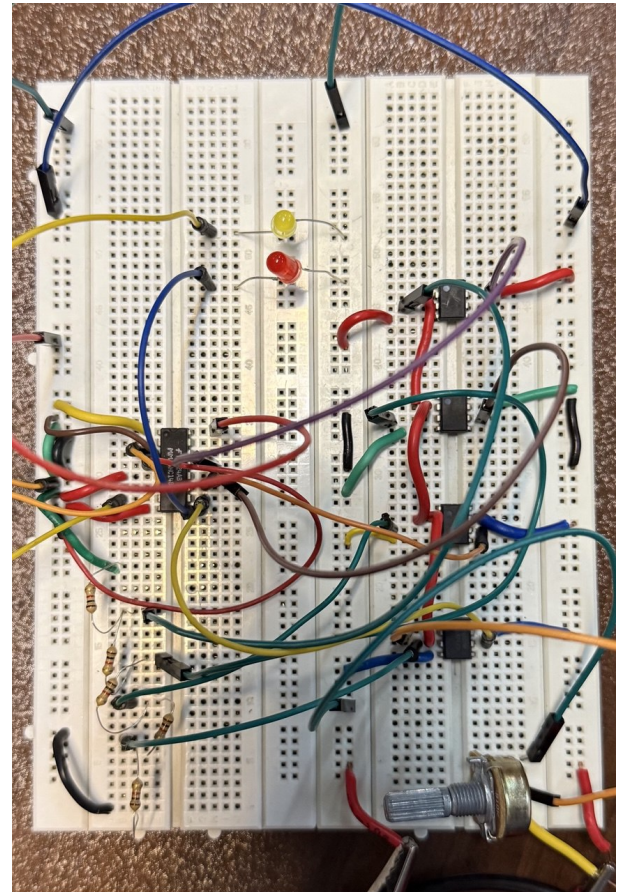


Fig. 4. View of the circuit with a different LED pattern, corresponding to another quantization region of the ADC.



Fig. 5. Measured digital waveform with a fundamental frequency of about 175.6 kHz.



Fig. 6. Digital output at a lower frequency of about 64.9 kHz.



Fig. 7. Output waveform at approximately 1.35 MHz. Ringing and overshoot are visible around the switching edges.

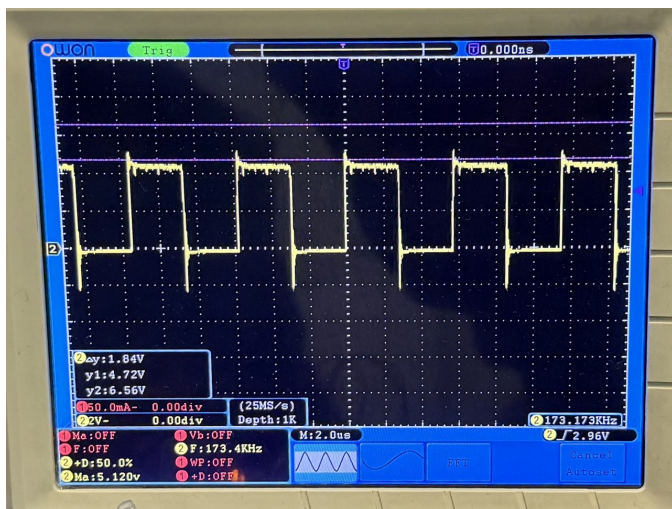


Fig. 8. Digital waveform at a frequency of about 173 kHz.